Cayley Graph Quotients and Twinless Paths

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1 Background

- Group Actions

  Let G be a group with, and let X be a set. A *(left) group action* of G on X is a map

  $$\cdot : G \times X \to X,$$

  usually written $(g, x) \mapsto g \cdot x$, such that for all $g, h \in G$ and all $x \in X$: $(gh) \cdot x = g \cdot (h \cdot x)$. In this project, we considered these actions on their corresponding Cayley Graphs, along with specific actions such as the monodromy and deck action.

- Cayley Graphs

  The *Cayley graph* $\mathrm{Cay}(G, T)$ of a group G with respect to a generating set T is the directed graph with vertex set G and edge set

  $$\{(g, gt) \in G \times G : g \in G \text{ and } t \in T\} = \big\{(g, h) \in G \times G : g^{-1}h \in T\big\}.$$

  That is, two vertices are connected by an edge if they differ by a generator.

  A group G acts on its Cayley graph $\mathrm{Cay}(G, T)$ by left multiplication. For example, the quotients

  $$A\backslash\mathrm{Cay}(G,T) \qquad\qquad B\backslash\mathrm{Cay}(G,T)$$

- Fundamental Group of Surfaces

  Let X be a topological space and $x_0 \in X$. Let $\pi_1(X, x_0)$ denote the set of loops based at x_0 up to a continuous transformation called a homotopy. We define the fundamental group as these loops based at x_0 with a group operation of concatenation.

- Cayley Graph Morphisms

  The action of G on itself by left multiplication extends to an action of G on $\mathrm{Cay}(G, T)$ by color-preserving graph automorphisms (colored by the generators). For each subgroup $H \leq G$, we denote

  $$\mathrm{Cay}(G, T; H) = H\backslash\mathrm{Cay}(G,T)$$

  to be the quotient associated to the restriction of this action to one of H on $\mathrm{Cay}(G, T)$.

- Almost Conjugate Subgroups

  Two subgroups A and B of a finite group G are called *almost conjugate* if $|A \cap C| = |B \cap C|$ for every conjugacy class C in G. We are in particular interested in almost conjugate subgroups that are not themselves conjugate.

- Twinless Paths

  In this project we defined a twinless path with a path in the form of $a^m b^n$ where a and b are generators of your group. One can think of this as following an "a" colored edge m times then following a "b" colored edges n times.

- Lifting Properties and Covering Spaces

  $p : \tilde{X} \to X$ is a covering map between topological spaces if for every $x \in X$, there is some neighborhood $U \ni x$ such that $p^{-1}(U)$ is a disjoint union of sets each homeomorphic to U (for example, the real line is a covering space of S^1 via the map $t \to \sin(2\pi t)$)

- Simple Lifts

  A key property of a covering space is that any continuous path on the base space can be lifted to a unique continuous path of the covering space. We say that a loop on the base space is simple if it can be lifted to an equivalence class that contains an injective (non-self-intersecting) loop.

- Finite Degree Covers and Finite Groups

  We can send the fundamental group into a finite group via a surjective homorphims by sending generating elements to generating elements of G and the rest to the identity in the case of left over generators. By fixing two finite degree based covers $p\colon (X, x) \to (S, s)$ and $q\colon (Y, y) \to (S, s)$. Let $K \trianglelefteq \pi_1(S, s)$ be a finite index normal subgroup of $\pi_1(S, s)$ which is contained in both $p_*\pi_1(X, x)$ and $q_*\pi_1(Y, y)$; for example, one may take K to be the normal core

  $$K = \bigcap_{\gamma \in \Gamma} \gamma(p_*\pi_1(X, x) \cap q_*\pi_1(Y, y))\gamma^{-1}$$

  of $p_*\pi_1(X, x) \cap q_*\pi_1(Y, y)$ in $\pi_1(S, s)$. Denote by $G = \dfrac{\pi_1(S, s)}{K}$ the corresponding quotient of $\pi_1(S, s)$. The quotient homomorphism $\rho\colon \pi_1(S, s) \twoheadrightarrow G$ is a surjective homomorphism of $\pi_1(S, s)$ onto a finite group G which takes the subgroups $p_*\pi_1(X, x)$ and $q_*\pi_1(Y, y)$ onto subgroups

  $$A = \rho(p_*\pi_1(X, x)) = \frac{p_*\pi_1(X, x)}{K} \qquad B = \rho(q_*\pi_1(Y, y)) = \frac{q_*\pi_1(Y, y)}{K}$$

  of G, respectively.

2 The Problem

Many questions in geometry, topology, and group theory reduce to how a group acts on a space. A particularly interesting setting arises when a finite group G acts by deck transformations on a covering space. In this context, each subgroup $A \leq G$ corresponds (via the Galois correspondence)

to a finite cover a surface, and the geometry of the cover is reflected in the combinatorics of the quotient graph $A\backslash\mathrm{Cay}(G,T)$.

Cayley graphs provide a natural setting for analyzing pairs of almost conjugate subgroups. Given a generating set $T \subseteq G$, the quotient graphs $A\backslash\mathrm{Cay}(G,T)$ and $B\backslash\mathrm{Cay}(G,T)$ encode how loops on a base surface lift to its corresponding covers. In particular, counting lifts of certain structured paths (for example, twinless paths of the form $s^m t^n$) offers a unique method for detecting geometric distinctions between the covers corresponding to A and B.

Our question is the following: given two almost conjugate (but not actually conjugate) subgroups A and B of a finite group G, can we distinguish Cayley graph quotients by counting *simple twinless paths*?

The reason for this is that if we are able to find almost conjugate but not conjugate subgroups $A, B \leq G$, then we can use these subgroups to generate length isospectral but not isometric measures on a finitely-covered base by two closed surfaces with negative Euler characteristic.

3 Problems along the Way

One of the major difficulties we had is that almost conjugacy (or Gassmann Equivalence) is not a very common property; SageMath did not provide libraries to calculate the property efficiently. As such, we had to code many of our functions by hand.

Additionally, many features such as conjugacy classes and subgroups are not compatible with groups other than the symmetric groups, like semi-direct products or matrix groups, so we had to limit our search to subgroups of symmetric groups (though by Cayley's theorem, every group is isomorphic to a subgroup of a symmetric group, so generality is not lost).

Another problem is optimality of the functions: due to the large nature of symmetric subgroups, our algorithm to discover almost conjugate subgroups had to brute force check all same-sized subgroups against each other. For the smaller symmetric groups, such as S7, this search took minutes. For larger symmetric groups, such as S10, this process took days. A bright side, however, is that our methods of distinguishing subgroups are easily parallelizable. In the future, we will be able to run checks faster and in a simultaneous manner.

4 Summary of Progress

Computational Search in Symmetric Groups. We performed a search for almost conjugate subgroup pairs in the symmetric groups $S_6, S_7, S_8, S_9,$ and S_{10}. For each S_n, we enumerated representatives of all conjugacy classes of subgroups, grouped them by order, and compared their conjugacy.

Discovered Almost Conjugate Subgroups. From S_6 to S_{10} we found many pairs of almost conjugate subgroups, one such example is the following subgroups in S_{10}:

$$A = \langle\ (1,4)(5,7), (1,7)(4,5), (2,3)\ \rangle, \qquad B = \langle\ (4,5)(7,9), (2,3), (1,10)(4,5)\ \rangle.$$

Simple Twinless Lift Criteria (Unlinked Case). Given generators $s, t \in T$ and integers m, n, a path of type $s^m t^n$ starting at the vertex Ag in $A\backslash\mathrm{Cay}(G,T)$ has a simple closed lift when:

$$\begin{aligned}
&g^{-1}s^j g \notin A && (1 \le j < |m|),\\
&g^{-1}s^{-m}t^k s^m g \notin A && (1 \le k < |n|),\\
&g^{-1}s^m g \in A \Rightarrow mn > 0,\\
&g^{-1}s^m t^n g \in A.
\end{aligned}$$

These criteria are implemented in our method `is_simple_lift_unlinked`.
Work so Far. We've so far implemented methods to:

- count simple twinless lifts of paths $s^m t^n$ in $A\backslash G$,
- compare lift counts for Almost Conjugate pairs (A, B),
- test whether (A, B) are distinguishable via unlinked simple lifts.

Implementation. We implemented our search in SageMath, using its native interface with GAP to work with the large subgroups of the aforementioned symmetric groups. For each pair of almost conjugate subgroups A and B of a finite group G, our code systematically checks whether the Cayley graph quotients $A\backslash\mathrm{Cay}(G,T)$ and $B\backslash\mathrm{Cay}(G,T)$ can be distinguished by counting simple twinless paths of type $s^m t^n$. The unlinked case is completely implemented and uses four criteria. The structure of the code and link to the GitHub follows:

```
distinguishable_overall(G, A, B, T)
│
├── distinguishable_unlinked(G, A, B, s, t)
│     └── loop over integers m, n with |m|,|n| ≤ [G:A]
│         └── distinguish_simple_lifts_unlinked(G, A, B, s, t, m, n)
│             ├── count_simple_lifts_unlinked(G, A, s, t, m, n)
│             │     └── iterate over all g in G
│             │         └── is_simple_lift_unlinked(G, A, s, t, m, n, g)
│             └── count_simple_lifts_unlinked(G, B, s, t, m, n)
│                   └── iterate over all g in G
│                       └── is_simple_lift_unlinked(G, B, s, t, m, n, g)
│
└── distinguishable_linked(G, A, B, s, t)
      └── loop over integers m, n with |m|,|n| ≤ [G:A]
          └── distinguish_simple_lifts_linked(G, A, B, s, t, m, n)
              ├── count_simple_lifts_linked(G, A, s, t, m, n)
              │     └── iterate over all g in G
              │         └── is_simple_lift_linked(G, A, s, t, m, n, g)
              └── count_simple_lifts_linked(G, B, s, t, m, n)
                    └── iterate over all g in G
                        └── is_simple_lift_linked(G, B, s, t, m, n, g)
```

Figure 1: Method Hierarchy

Figure 2: QR code to Github repository

5 Future Direction/Continuation

Simple Twinless Lift Criteria (Linked Case). In the future, we will implement the linked case, which has the following criteria:

$$\begin{aligned}
&gs^m t^n g^{-1} \in A\\
&gs^m t^k s^{-j} g^{-1} \notin A && (1 \le k < |n|), (1 < j < m)\\
&g^{-1}s^m g \in A \Rightarrow mn > 0,
\end{aligned}$$

this will be implemented in our methods with the linked chain of commands. Which we will compare to the unlinked case to see whether the the subgroups are distinguishable given the Group and a generating set.

6 References

1 Aougab, Lahn, Loving, Miller (2025) Unmarked simple length spectral rigidity for covers. Mathematische Annalen. https://doi.org/10.1007/s00208-025-03103-y

2 Lahn (2025) Cayley Graph Quotients and Twinless Paths, Typed Notes

3 Sage Math Documentation. https://doc.sagemath.org/html/en/reference/